



# Non-uniform and volumetric effect on the hydrodynamic and thermal characteristic in a unit solar absorber

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## ARTICLE INFO

### Article history:

Received 10 October 2020

Received in revised form

17 January 2021

Accepted 15 February 2021

Available online 18 February 2021

### Keywords:

Volumetric effect

Hydrodynamic characteristic

Solar absorber

Porous medium

Thermal performance

## ABSTRACT

Experimental and numerical analyses on the thermal and hydrodynamic performance of volumetric solar receivers were conducted. Silicon carbide (SiC) absorbers with various pore structures were systematically investigated on a laboratory-scale test platform. A three-dimensional theoretical model coupling the fluid flow with internal heat transfer was presented. The major characteristics of the thermal and hydrodynamic behavior combining radiation, thermal conduction, and interphase convection in the absorber were presented. In addition, the key design parameters were systematically analyzed. The experimental results revealed that a small pore diameter significantly enhances the volumetric heat-transfer coefficient, which in turn increases the fluid temperature and subsequently leads to a high thermal efficiency. Under high local thermal non-equilibrium state, the non-uniform hydrodynamic characteristic significantly influenced due to the increase in the fluid viscosity at the hot spot, especially in the case of a small-pore-sized absorber, thereby increasing the risk of overheating. For absorbers with similar porosities ( $\phi \sim 0.85$ ),  $d_p$  decreased from 5.83 mm to 2.22 mm, and the thermal efficiency  $\eta$  increased by nearly 3%. Moreover, the simulation results revealed that the absorber unit with  $d_p$  of 2.22 mm and  $\phi$  of 0.95 exhibits the maximum thermal efficiency of 72.48%.

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## 1. Introduction

Solar porous volumetric receivers demonstrate promise for solar thermal utilization due to their capabilities of achieving high temperature and thermal efficiency [1]. In recent years, solar thermochemical reactor technology that produces substances such as metals, fresh water, hydrogen, and fuels [2] by the heat derived from concentrated solar irradiation has attracted considerable attention. On the other hand, the thermodynamic cycle, such as an open-loop air system with the Rankine cycle and a closed-loop system with the Brayton cycle, still demonstrates promise as a technology for large-scale commercial applications in solar thermal power plants [3,4]. Typically, porous volumetric receivers comprise porous materials such as metal wires, ceramic foam, fiber mesh, and catalyst carriers, etc., which exhibit high temperature resistance, a high specific surface area, and a tortuous flow channel [5,6]. These characteristics make porous volumetric receivers suitable for thermochemical reactions and thermodynamic cycles that require

high temperature and fluid mixed.

In the past decades, dozens of solar volumetric receiver prototypes have been designed and developed on an experimental scale of several kilowatts to the application scale of megawatts [7]. Various potential materials for volumetric receivers, such as metal wires, ceramic foam, fiber mesh, and catalyst carriers, have been designed and tested. Fend et al. [8,9] have conducted experiments on the evaluation of thermophysical and heat-transfer properties of several porous materials such as open volumetric solar receivers. They concluded that materials based on ceramic foams or fabrics would be promising candidates due to their high specific surface area and low-pressure loss. Pabst et al. [10] have proposed an advanced metal honeycomb receiver and investigated a 500-kW laboratory-scale research platform. Results revealed that a high cellularity and high porosity positively affect thermal efficiency. The best efficiency has been reported for a high-cell-density material with a porosity of greater than 90%. Recently, Avila-Marin et al. have proposed and investigated gradual porosity volumetric receivers [11] to explore the scheme for deeper solar flux penetration and thus reduce the temperature gradient. The results revealed that the structure with the highest possible porosity at the

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entrance and the highest possible specific surface area at the exit exhibits better thermal behavior. Similarly, Zaversky et al. [12] have conducted a series of experimental and numerical studies on ceramic foams as a solar absorber material with single- and multi-layer configuration. Their results revealed that foam porosity is positively correlated with thermal efficiency. The characteristics of the first layer apparently determine the thermal performance of the receiver as a significant correlation cannot be observed between the characteristics of the second or third layer and thermal efficiency. Unlike traditional porous media materials, Luque et al. [13] have proposed a novel hierarchically layered fractal-like volumetric receiver with stainless steel by additive manufacturing technology. The volumetric effect is detected by experimental tests, but the efficiency is not considerably different from that of a conventional porous media absorber, possibly related to the high reflectivity of stainless steel and the low convective heat-transfer rates. From the above-mentioned studies, materials with high porosity or high cellularity positively affect the thermal performance of a volumetric receiver. High porosity is linked to a low extinction coefficient, which in turn reduces the radiation flux gradient. On the other hand, high cellularity is linked to a high specific surface area, which provides efficient heat-transfer conditions [11].

To better understand the mechanism of heat and mass transfer in solar porous volumetric receivers, several studies have focused on building a numerical model for describing transport phenomena inside porous media. There are two approaches employed to model the transport process inside the receivers: one is micro-modeling based on a pore-level approach that combines 3D digital porous geometry; one is macro-modeling based on a continuous-scale approach with the hypothesis that considers the porous receiver as a continuous semi-transparent medium. In terms of the pore-level approach, the complex geometry of the porous materials is modeled, and the flow and heat-transfer process inside the pores are simulated on the basis of the computational fluid dynamics method [14]. In this manner, transport phenomena inside the pores are well displayed and show better consistency with the experimental results. However, this approach involves complicated grid processing and high computation costs. In several engineering applications and research, it is not necessary to completely restore pore-level phenomena, and researchers are more concerned about the indicators of the macro-phenomena. Hence, the continuous-scale approach is the tool of choice for several researchers due to its low computation costs as well as convenience in coupling the multiphysics, such as chemical reactions. In this regard, various studies have aimed to develop an empirical model to describe the heat transfer characteristics and flow resistance in porous media. Experimental and numerical studies of flow and heat transfer characteristics of porous media have been reported by Dietrich [15], Wu [16], Regulski [17], P.X. Jiang [18], various empirical correlations for pressure drop prediction have been proposed. In addition, flow resistance characteristics of metal porous materials have been investigated by several researchers, such as copper foams [19,20], aluminum foams [21], wire mesh [22], and sintered bronze particle [23,24] etc.

For modeling the main heat-transfer processes inside a porous receiver, the local thermal non-equilibrium (LTNE) model is widely employed in several studies. In the LTNE model, the temperature inside the porous receiver is divided into two parts, solid and fluid, respectively, and two energy equations are needed to represent the solid and fluid phases, respectively. As forced convection is considered as the main heat-transfer process between the solid and fluid phases, for a more accurate prediction of these temperature phases, several studies have been devoted to the investigation of the convective heat transfer inside porous materials. For instance, Avila-Marín et al. [22] have conducted a numerical study to obtain

the convective heat-transfer coefficient between the air flow and a plain-weave wire mesh. Wu et al. [25] have conducted a numerical study of the convective transfer between the air flow and ceramic foams by modeling the pore structures as an idealized packed tetrakaidecahedron and proposed a correlation for the local convective heat-transfer coefficient. Dietrich [26] has developed heat-transfer coefficients for various ceramic materials by transient experiments.

It is crucial to investigate the radiative heat-transfer process inside pores for the study of the thermal performance of volumetric receivers. Radiative properties of porous media are closely related to pore structure characteristics, such as porosity, cell diameter, strut diameter, and cross-sectional shape of strut [27–30]. Li et al. have reported the latest analytical relations for predicting the extinction coefficients, scattering albedos, and asymmetry factors of irregular open cell foams [31,32].

Typically, the radiation process inside a porous receiver is assumed to comprise two parts [33,34], one is the transfer of incident irradiation, referring to the rays scattering or absorption inside the porous medium; and the other is the transfer of thermal radiation, referring to the emission of thermal irradiation by the inner high-temperature porous structure. For modeling the incident irradiation transfer process, the Monte Carlo ray tracing (MCRT) method based on probability statistics has been well established by several researchers [35]. In this method, the porous structure is considered as a semitransparent medium [29,36], the propagation of each ray is subject to the extinction and scattering characteristic of a porous medium, and all trajectories are tracked until the ray is absorbed or leaves the medium. Further development has been reported by Barreto et al. [37], in which the Henyey–Greenstein phase function is embedded into the ray tracing model for porous media to solve the problem of biased scattering. To solve the thermal radiation transfer inside the porous receiver, several methods have been established, such as  $P_N$  approximation,  $S_N$  approximation (DO method), and the MCRT method [36]. Usually, the P-1 approximation is widely employed due to its accuracy and convenience to couple with the energy equation. Wang et al. [38–40] have well verified the applicability of the P-1 approximation and its numerical validation on the radiation transfer inside the porous receiver. Chen et al. [33] and Barreto et al. [34] have applied the P-1 model to address the radiative field and coupled the radiation equation with the energy equation of a solid via the radiation source term.

The extremely high temperature gradients inside the solar absorber due to the concentrating irradiation of the dish system is the key limitations of solar absorber and as well as the thermal reactor. However, from the literature reviewer we find that little attention has been devoted to an overall analysis of non-uniform and volumetric effect on the hydrodynamic and thermal characteristic in a unit solar absorber. In this study, four typical porous silicon carbide absorbers were examined on a solar lamp experiment platform to obtain the real non-uniform temperature distribution of the porous absorber. Meanwhile, a three-dimensional theoretical model adequately coupling the solar radiation transport and internal heat transfer is established. To quantitative analysis the major characteristics of the thermal and hydrodynamic behavior combining radiation, thermal conduction, and interphase convection in the absorber.

## 2. Experimental setup

For stable testing without being affected by the weather, an indoor receiver test platform is established (Fig. 1). The aim of the experiment project is to examine the thermal and hydrodynamic behavior of solar porous volumetric receivers and to establish the

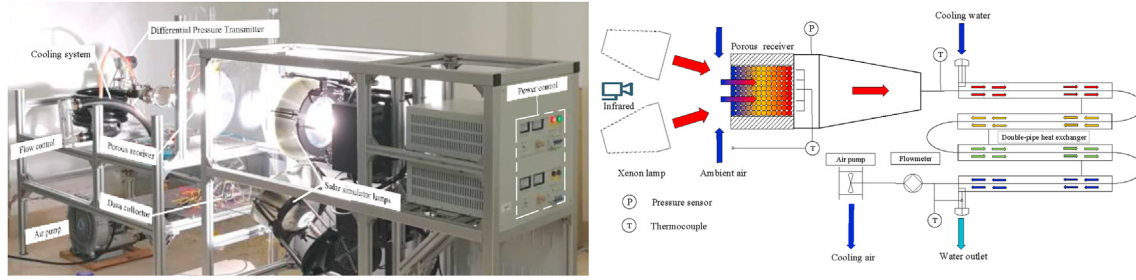


Fig. 1. Schematic of the test platform and sensor arrangement.

numerical model. The test platform mainly comprises five sub-systems, i.e., solar simulator, porous receiver, air circulation system, water cooling system, and data acquisition system, respectively. The radiation power emitted by the solar simulator can reach up to 2.0 kW in a few seconds, and the peak radiative flux density is  $\sim 1000 \text{ kW/m}^2$ . The solar flux distribution profiles on the receiver surface are measured by a solar flux measurement system, which is widely used for measuring the distribution of concentrated solar flux [41].

In the experimental process, the solar simulator stably emits kilowatts of concentrated light at the entrance of the receiver, and then the air flows into the receiver and carries the heat away by forced convection. Later, the high-temperature air is cooled to the level of ambient temperature by using a sleeve water-cooled heat exchanger before it is released into the environment. At the start up, the air flow rate is set to the maximum to prevent overheating. Then, the air flow rate is artificially controlled to achieve various air outlet temperatures. Two infrared thermometers are used to measure the surface temperature of the receiver. Several thermocouples are distributed at different positions of the test platform to obtain various temperatures. All sensors are integrated into LabVIEW software and recorded in real time. Fig. 2 shows the SiC foam samples and the temperature distribution images of the irradiated surface captured by an infrared camera. The decrease in the pore diameter  $d_p$  and porosity  $\phi$  leads to the gradual shrinking of the

high-temperature area. This phenomenon is closely related to the effect of the pore structure on the hydrodynamic and thermal volumetric distributions in the SiC foam absorber unit, which is systematically discussed in the following part. More detailed experimental data can be obtained from our previous study [42].

### 3. Mathematical model

#### 3.1. LTNE model

Continuum equation:

$$\nabla(\rho_f \mathbf{u}) = 0 \quad (1)$$

Momentum equation:

$$\frac{\rho_f}{\phi} \left( (\mathbf{u} \cdot \nabla) \frac{\mathbf{u}}{\phi} \right) = -\nabla p + \nabla \cdot \left( \frac{\mu_f}{\phi} (\nabla \mathbf{u} + (\nabla \mathbf{u})^T) \right) - \frac{2\mu_f}{3\phi} (\nabla \cdot \mathbf{u}) - \frac{\mu_f}{K} \cdot \mathbf{u} + \mathbf{F} \quad (2)$$

where the  $\mathbf{u}$  is the velocity vector of the fluid,  $\rho_f$  is the fluid density,  $\mu_f$  is the fluid dynamic viscosity, and  $\phi$  is the porosity. The permeability tensor  $K$  and the flow resistance  $\mathbf{F}$  caused by the porous media can be calculated by the following correlation, respectively

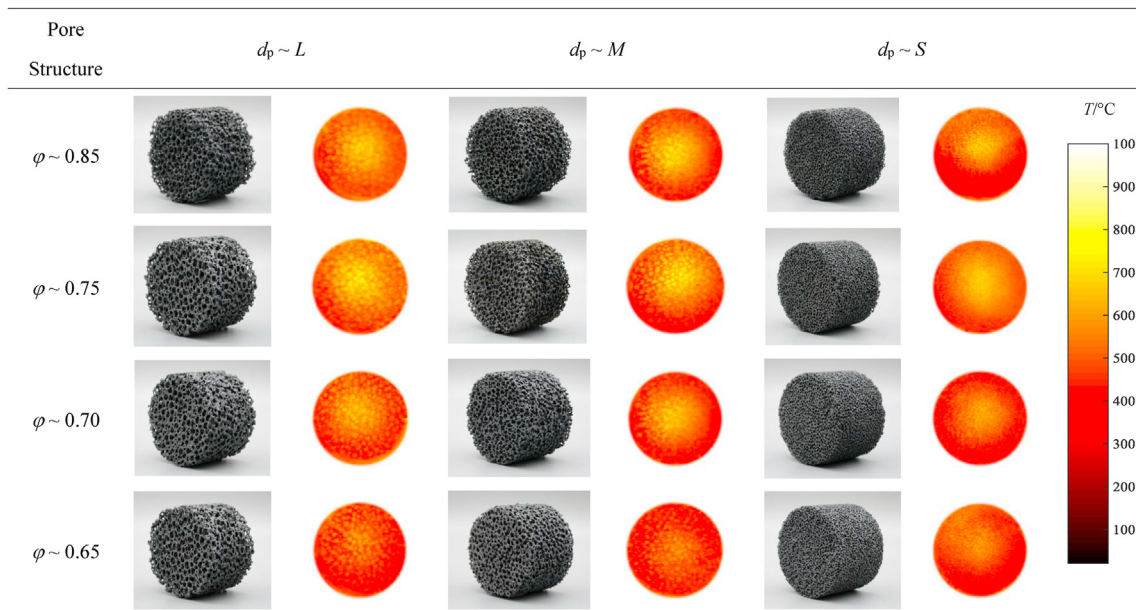


Fig. 2. SiC foam samples and the temperature distribution image of the irradiated surface captured by an infrared camera.

[16].

$$K = \frac{\phi^3 d_p^2}{150(1-\phi)^2} \quad (3)$$

$$F = -\frac{1039 - 1002\phi}{d_p^2} \mu_f \mathbf{u} - \frac{0.5138\phi^{-5.739}}{d_p^2} \rho_f |\mathbf{u}| \mathbf{u} \quad (4)$$

where  $d_p$  is the pore diameter. This flow resistance correlation is valid for the flow condition with the Reynolds number in the range of  $10 < Re < 400$  and foam porosity in the range of  $0.66 < \phi < 0.93$ , which are suitable for the working condition of the experimental and simulation models in this study.

Energy equations:

Fluid phases:

$$\rho_f C_p \mathbf{u} \cdot \nabla \langle T_f \rangle = \nabla \cdot (\lambda_{fe} \nabla \langle T_f \rangle) + h_{sf} (\langle T_s \rangle - \langle T_f \rangle) \quad (5)$$

Solid phases:

$$0 = \nabla \cdot (\lambda_{se} \nabla \langle T_s \rangle) + h_{sf} (\langle T_f \rangle - \langle T_s \rangle) + Q_{rad} + Q_{inc} \quad (6)$$

where  $\langle T_f \rangle$  and  $\langle T_s \rangle$  are phase temperatures of the fluid and solid, respectively, and  $C_p$  is the specific heat of the fluid. Here, the radiative source term is divided into two parts:  $Q_{rad}$  is the source term caused by the thermal radiation transfer inside the porous structure, and  $Q_{inc}$  is the heat source term caused by the incident solar radiation absorption. The effective thermal conductivity of the fluid phase  $\lambda_{fe}$  and solid phase  $\lambda_{se}$  can be expressed as follows, respectively:

$$\lambda_{fe} = \phi \lambda_f \quad (7)$$

$$\lambda_{se} = (1 - \phi) \lambda_s / 3 \quad (8)$$

The thermal conductivity of the solid (silicon carbide)  $\lambda_s$  is considered to be a constant at 80 W/m/K. The thermal conductivity of the fluid (air)  $\lambda_f$  is dependent on the temperature, which is defined as a function of the temperature that fits the data obtained from the study of Mulholland [43].

The volumetric heat-transfer coefficient  $h_{sf}$  is the key parameter that significantly affects the numerical prediction of the heat transfer inside the porous receiver. As the foam samples used in this study exhibit high porosity and a large pore size, and considering the low Reynolds number in this experiment, empirical correlations established by Younis et al. [44] have been used:

$$h_{sf} = \left( 0.819(1 - 7.33(d_p/L)) \text{Re}^{0.36(1+15.5(d_p/L))} \right) \lambda_f / d_p^2 \quad (9)$$

The scope of this empirical formula is  $0.005 < d_p/L < 0.136$  and  $5.1 < Re < 564$ . The selection of experimental formulas is discussed in detail in Section 5.2.

Chen [33] has reported that a radiation process is assumed to comprise two parts: incident irradiation transfer and thermal radiation transfer. The former refers to the scattering or absorption of rays in the porous medium, while the latter refers to the thermal radiative exchange inside the high-temperature porous structure.

For the thermal radiation transfer of the porous receiver, a modified differential approximation (P-1 Model) has been applied to address the problem. Similar to the classic P-1 model, thermal radiation is treated to be fairly diffuse, which is the result of emissions within the porous medium. As the external irradiation transfer is removed from the intensity field, the remnant intensity can only slightly deviate from the isotropic condition. The radiative

transport equation inside the porous medium can be expressed as follows:

$$0 = \nabla \cdot \left( \frac{1}{3\beta} \nabla G_d \right) + \kappa_s (4\sigma \langle T_s \rangle^4 - G_d) \quad (10)$$

where  $G_d$  is the incident radiation, and  $\sigma$  is the Stefan Boltzmann constant. Then, the coupling source term of radiative heat transfer and solid heat conduction can be obtained:

$$Q_{rad} = -\kappa_s (4\sigma \langle T_s \rangle^4 - G_d) \quad (11)$$

To solve the irradiation process inside the porous absorber, the MCRT method was developed, which is widely used in modelling the solar radiation absorption in a porous volumetric receiver. In this case, the porous structure is considered as a semitransparent medium, the propagation of each ray is subject to extinction and scattering characteristics of the porous medium, and all trajectories are tracked until the ray is absorbed or leaves the medium. The absorption coefficient  $\kappa$ , scattering coefficient  $\sigma$ , and extinction coefficient  $\beta$  can be expressed as follows [38]:

$$\kappa_s = \frac{\Psi}{2d_p} \epsilon (1 - \phi) \quad (12)$$

$$\sigma_s = \frac{\Psi}{2d_p} (2 - \epsilon) (1 - \phi) \quad (13)$$

$$\beta = \kappa_s + \sigma_s = \frac{\Psi}{d_p} (1 - \phi) \quad (14)$$

where the  $\epsilon$  is the emissivity of porous materials.  $\Psi$  is an empirical coefficient based on the properties of reticulated porous ceramic (RPC) [45]. When concentrated light is transmitted into the porous receiver, the free path of the ray can be obtained as follows:

$$l = -\frac{1}{\beta} \ln \Gamma \quad (15)$$

where the random number  $0 < \Gamma < 1$  is generated to determine the free path of the ray. When the position of a ray is established, another random number  $\xi$  ( $0-1$ ) is used to determine whether the ray will be scattered ( $\xi > \omega$ ) or absorbed ( $\xi \leq \omega$ );  $\omega$  is the scattering albedo ( $\omega = \sigma/\beta$ ). When scattering occurs, new directions are calculated by spherical coordinates, which are expressed in the polar angle  $\theta$  and the azimuth angle  $\psi$ .

$$\psi = 2\pi\xi \quad (16)$$

$$\theta = \cos^{-1}(2\xi - 1) \quad (17)$$

After completing the tracing process of all incident rays and determining the final position of each ray, the spatial distribution of the absorbed solar radiation inside the porous receiver can be obtained:

$$Q_{inc} = \frac{N_{i,j,k}^S \cdot e}{V_{i,j,k}} \quad (18)$$

where  $N^S$  is the statistical ray's number in the spatial grid,  $e$  is the energy of each ray, and  $V$  is the volume of the local grid. The fraction of scattering loss from the irradiated surface ( $\eta_s$ ) and the fraction of energy absorbed by the porous receiver ( $\eta_o$ ) are obtained.



**Table 1**  
Physical parameters of SiC ceramic foams.

| Parameter                                          | Value      |
|----------------------------------------------------|------------|
| Radius of porous receiver $R_{rec}$                | 35.0 mm    |
| Radius of insulation layer $R_{inc}$               | 100.0 mm   |
| Thickness of porous foam $L$                       | 50.0 mm    |
| Thermal conductivity of insulation $\lambda_{ins}$ | 0.16 W/m/K |
| Thermal conductivity of solid (SiC) $\lambda_s$    | 80.0 W/m/K |
| Emissivity of solid $\epsilon_s$                   | 0.92       |

$$\eta_s = \frac{N_{sca}}{N_{inc}} \quad (19)$$

$$\eta_o = \frac{N_{abs}}{N_{inc}} \quad (20)$$

where  $N_{inc}$ ,  $N_{sca}$ , and  $N_{abs}$  are the total number of the incident rays, the number of scattered rays from the irradiated surface, and the number of absorbed rays, respectively. In summary, the fraction of energy loss from scattering and the energy absorbed by the porous receiver are obtained.

$$q_{sac} = \eta_s \cdot p_{inc} \quad (21)$$

$$q_{abs} = \eta_o \cdot p_{inc} \quad (22)$$

### 3.2. Boundary conditions

As the ambient air is drawn into the receiver by a high-pressure pump, the inlet air temperature is considered to be equal to the ambient temperature, and the air flow rate is determined; thus, the flow boundary conditions of the receiver are expressed as follows:

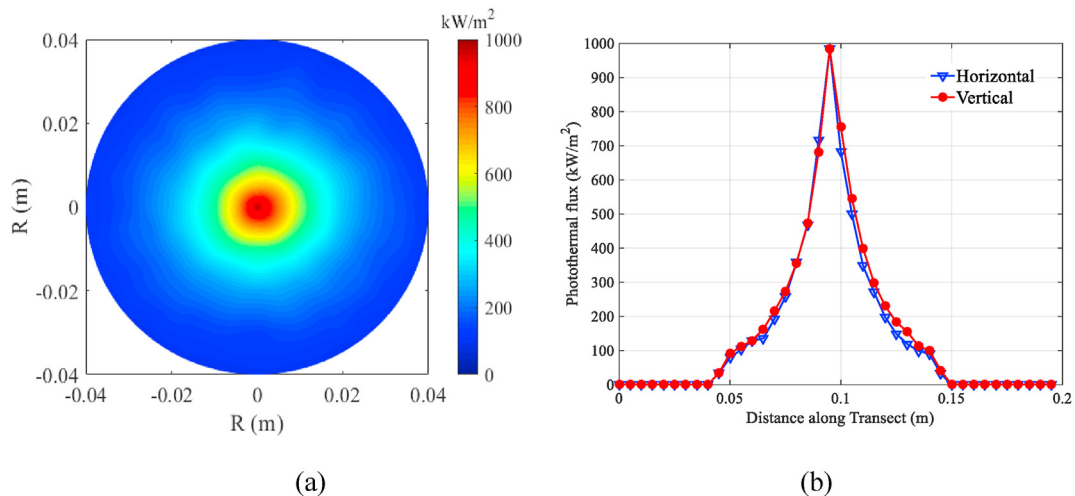
$$\begin{aligned} Z = 0 : \langle T_f \rangle &= T_e, \quad m_f = \int_A \rho_f \mathbf{u} ds, \quad \nabla p = 0 \\ Z = L : \langle T_f \rangle &= 0, \quad \nabla \mathbf{u} = 0, \quad p = p_{out} \end{aligned} \quad (23)$$

where the outlet pressure is set at a constant pressure,  $A$  is the cross-sectional area of the receiver, and  $m_f$  is the air flow rate.

In the receiver inlet, as the air flow is drawn into the receiver, it can be considered that there are no convective and conduction heat losses from the irradiated surface of the receiver. The major heat loss from the irradiated surface includes thermal radiation. For the P1 radiation model, Marshak boundary conditions are applied to the solid boundary. The lamp is considered to be sufficiently far from the irradiated surface such that the effect of mutual radiation is ignored, i.e., the thermal radiation of the porous receiver is directly emitted into the environment. Thus, diffuse radiation and conductive heat transfer on the irradiated surface can be expressed as follows:

**Table 2**  
Comparison between the experiment and simulation results. (Incident heat flux power: 1240 W).

| Case ID | $\phi$ | $d_p$ (mm) | $M_v$ (m <sup>3</sup> /h) | $T_e$ (°C) | $T_{fo}$ exp. (°C) | $T_{fo}$ sim. (°C) | Error abs.(°C) | Temp deviation (%) |
|---------|--------|------------|---------------------------|------------|--------------------|--------------------|----------------|--------------------|
| L-1     | 0.852  | 5.83       | 4.0                       | 28.17      | 459.50             | 468.32             | 8.82           | 1.92%              |
|         |        |            | 8.0                       | 28.09      | 318.72             | 317.27             | 1.45           | 0.45%              |
|         |        |            | 12.0                      | 27.85      | 243.05             | 237.66             | 5.39           | 2.22%              |
| L-2     | 0.770  | 5.67       | 4.0                       | 27.88      | 484.56             | 473.30             | 11.26          | 2.32%              |
|         |        |            | 8.0                       | 27.91      | 337.30             | 321.11             | 16.19          | 4.80%              |
|         |        |            | 12.0                      | 28.08      | 263.81             | 241.56             | 22.25          | 8.43%              |
| S-1     | 0.844  | 2.22       | 4.0                       | 27.75      | 480.51             | 525.73             | 45.22          | 9.41%              |
|         |        |            | 8.0                       | 29.38      | 340.14             | 333.26             | 6.88           | 2.02%              |
|         |        |            | 12.0                      | 29.03      | 252.48             | 245.89             | 6.59           | 2.61%              |
| S-2     | 0.754  | 2.30       | 4.0                       | 28.41      | 492.13             | 534.63             | 42.5           | 8.64%              |
|         |        |            | 8.0                       | 27.94      | 353.58             | 336.35             | 17.23          | 4.87%              |
|         |        |            | 12.0                      | 27.72      | 272.47             | 247.04             | 25.43          | 9.33%              |



**Fig. 3.** Solar radiative flux distribution on the receiver surface (a) and flux distribution along the vertical and horizontal transects at the peak value point (b).

$$\frac{1}{3\beta} \nabla G_d|_{z=0} = \frac{G_d|_{z=0}}{2} \quad (24)$$

$$\lambda_{se} \nabla \langle T_s \rangle|_{z=0} = (1 - \phi) \epsilon \sigma (\langle T_s \rangle|_{z=0} - T_e^4) \quad (25)$$

where the right term of Eq. (24) represents the radiations emitting from the void space of the receiver. solid. The right term of Eq. (25) represents the radiosity  $J_s$  emitting from the solid, where according to the definition of porosity, the view factors between the void and solid to the environment are  $F_{v-e} = \phi$ ,  $F_{s-e} = 1 - \phi$ .

Unlike the irradiated surface, the outlet surface is close to the temperature measuring plate in this experiment, and the temperature of the measuring plate is assumed to be the same as the air temperature under stable conditions. Therefore, radiosity  $J$  and the net radiative heat flux  $q$  of each surface can be expressed as follows:

$$\frac{1}{3\beta} \nabla G_d|_{z=L} = \frac{\epsilon_1 (4\sigma \langle T_s \rangle^4)|_{z=L} - G_d|_{z=L}}{2(2 - \epsilon_1)} \quad (26)$$

$$-\lambda_{se} \nabla \langle T_s \rangle|_{z=0} = (1 - \phi) \epsilon_1 \sigma (\langle T_s \rangle^4 - \langle T_f \rangle^4) \quad (27)$$

where  $\langle T_f \rangle$  is the mean temperature of the outlet air temperature.

In the experiment, the side wall of the receiver is covered with a ceramic fiber insulation layer. During the experiment, the temperature of the outer surface of the insulation layer is almost the same as that of the environment; hence, convection between the outer surface and environment is considered sufficient, and the temperature of the outer surface is equal to the environmental temperature. At the interface between the solid and insulation

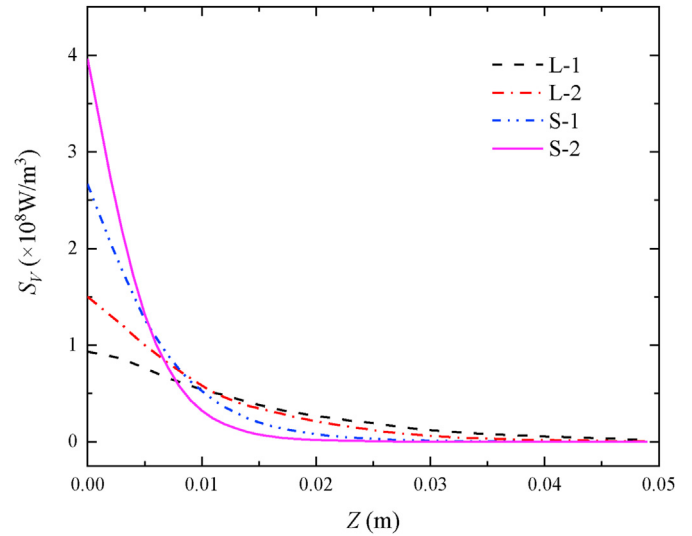


Fig. 5. Distribution of the solar radiative flux along the centerline of the receiver with four porous structures.

layer, the temperature of the fluid is assumed to be equal to that of the solid (no-slip condition). In this way, the boundary thermal transfer is simplified as radial thermal conduction, and the thermal resistance of thermal conduction can be expressed as follows:

$$-\lambda_{se} \frac{d\langle T_s \rangle}{dr}|_{r=R_{rec}} = \frac{\lambda_i}{\ln(R_{ins}/R_{rec}) R_{rec}} (\langle T_s \rangle|_{r=R_{rec}} - T_e) \quad (28)$$

where  $R_{rec}$  and  $R_{inc}$  are the radii of the porous receiver and

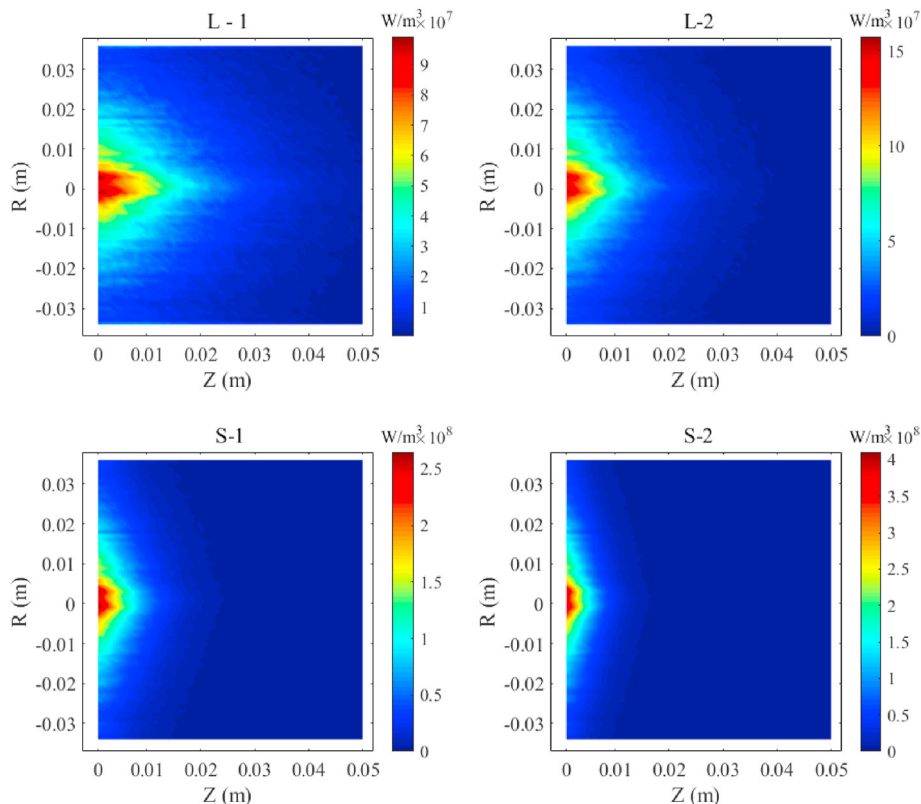


Fig. 4. Spatial distribution of the solar radiation absorption with four porous receivers.

insulation layer, respectively.  $\lambda_i$  is the thermal conductivity of the ceramic fiber insulation, which is set as a constant with 0.16 W/m/K.

#### 4. Numerical procedure and experimental validation

Solar radiation transport from xenon lamps to the interior of the porous receiver is established and solved by the in-house MCRT code. Detailed implementation and validation of the MCRT code can be seen from our previous work [46]. The total number of released rays in the MCRT code is set at  $10^9$  for an average deviation of less than 1% [37,47]. The spatial distribution of solar radiation absorption is employed as radiative source terms to couple the LTNE model with the P1 approximation. The applicability of the P1 approximation and its numerical validation on the radiation transfer has been well established in our previous study [38]. The governing equations for mass, momentum, and energy conservation are discretized using the SIMPLE algorithm via the application

of the finite element method [48,49]. The convective terms are discretized by employing an upwind differencing method. Convergence is considered to be achieved when the relative variation of velocity, pressure, and temperature between consecutive iterations is less than  $10^{-8}$  for all grid points in the computational domain after the grid independent test.

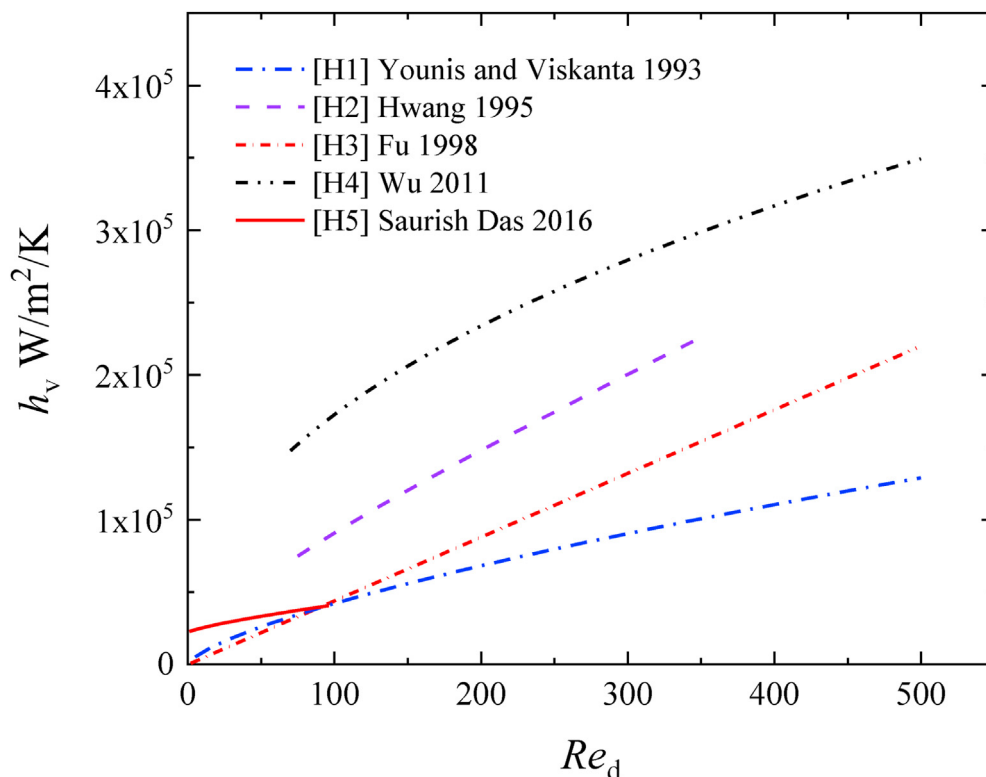
The thermal transfer model was validated by the comparison of the experimental data of measured temperatures with those obtained from numerical calculations. Four SiC foam receivers with various porosities and cell density are tested on a laboratory-scale volumetric solar receiver test platform. Table 1 lists the physical parameters of the SiC ceramic foams and experiment operating conditions, and Table 2 lists the comparison between experiment and simulation results. The simulation results under a series of flow rates for all cases are in good agreement with the experimental data, with the maximum deviation of the average outlet air temperature being only 9.41%.

**Table 3**

Correlations of the volumetric convection coefficient available in the literature

| Model             | $h_v/Nu_v$                                                                                                                                                                               | Limitation                                                            |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Younis 1993 [44]  | $Nu_v = 0.819(1 - 7.33(d_p/L))Re^{0.36(1+15.5(d_p/L))}$                                                                                                                                  | $0.005 < d_p/L < 0.136$ and $5.1 < Re < 564$                          |
| Hwang 1995 [50]   | $h_v = \left(1.064 \left(\frac{\lambda_f}{d_p}\right) Pr^{0.33} Re^{0.59} + 0.004 \left(\frac{d_v}{d_p}\right)^{0.35} \left(\frac{\lambda_f}{d_p}\right) Pr^{0.35} Re^{1.35}\right) / 2$ | $75 < Re < 350$                                                       |
| Fu 1998 [51]      | $Nu_v = \left(0.0426 + 1.236 \frac{d_m}{L}\right) Re_{dm,dm} = 4(\phi/\pi)^{0.5}/PPI$                                                                                                    | $0.742 < \phi < 0.916$ , $2 < Re_{dm} < 836$ , $0.031 < d_m/L < 0.26$ |
| [H4]Wu 2011 [25]  | $Nu_v = (32.504\phi^{0.38} - 109.94\phi^{1.38} + 166.65\phi^{2.38} + 86.98\phi^{3.38})Re$                                                                                                | $0.66 < \phi < 0.93$ , $70 < Re < 800$ ,                              |
| [H5]Das 2016 [52] | $Nu_v = 0.2 + 0.1583Re^{0.7796}Pr^{1/3}$                                                                                                                                                 | $1 < Re < 100$                                                        |

$$^a Nu_v = h_v d_p^2 / \lambda_f$$



**Fig. 6.** Volumetric heat-transfer coefficient versus the Reynolds number. (Pore structure:  $\phi = 0.8$ ,  $d_p = 3$  mm,  $L = 50$  mm,  $T = 400$  K).

## 5. Results and discussion

### 5.1. Optical absorption characteristics of porous receiver

As the thermal energy of the receiver is derived from solar radiation absorption, then the distribution characteristics of solar radiation absorption are directly related to the distribution of the temperature field of the receiver. According to the above-mentioned MCRT method, the absorption distribution of solar radiation in porous media can be approximately obtained. The solar radiative flux distribution on the receiver surface and flux distribution along the vertical and horizontal transect at the peak value point are shown in Fig. 3.

Fig. 4 shows the spatial distribution of the absorbed solar radiation of the receiver models used herein. The solar radiation absorption is mainly located at the receiver inlet within a few centimeters deep, and a cone-shaped spatial distribution of solar radiation absorption is observed due to the focus angle of the incident solar light. With the decrease in the porosity or pore size, the distribution area of the solar radiation absorption becomes narrow, and the peak value of solar radiation flux increases. Fig. 5 shows the solar radiative flux along the central line of those porous receivers. For the receivers with a high porosity or large pore size, the decline gradient of the solar radiative flux is gentle as the sparse structure reduces the extinction coefficient, resulting in deep radiation absorption and a low peak flux. The solar radiation absorption will be used as a heat source term in the energy equation to calculate the temperature field in a porous receiver.

### 5.2. Validation of the model using volumetric experiment test results

As a critical parameter that determines the heat-transfer rate between the solid and fluid phases, the volumetric heat-transfer coefficient  $h_v$  significantly affects the temperature distribution characteristics under LTNE conditions. However, owing to the complexity of the pore structure and diversity of material properties, most of the heat-transfer coefficient models available in literature cannot adapt to all situations; each of the models exhibit limitations, as detailed in Table 3. To find a heat-transfer coefficient model that is more consistent with the experimental results, some correlations are compared and discussed here; the most consistent one is adopted in our numerical simulation.

From Fig. 6,  $h_v$  calculated by the correlations is obtained, which is available in the range of our experiment, with a typical pore structure and temperature conditions ( $\phi = 0.8$ ,  $d_p = 3$  mm,  $L = 50$  mm,  $T = 400$  K). Under the condition of a low Reynolds number, most of the volumetric heat-transfer coefficients exhibit a linear change with the Reynolds number. Distinct differences are observed among the correlations; thus, the correlations must be carefully selected. Das [52] has reported that the correlation [H5] is suitable in the case of low Reynolds number, but lack data in case of high Reynolds numbers. Hwang et al. have reported a correlation [H2] where the Reynolds number is limited in a range of 75–350, while in some cases, the Reynolds number is greater than the maximum limit. Hence, according to the application range of Re, those correlations of Younis and Viskanta [1], Fu [H3], and Wu [H4] may be applicable. Then, these correlations are applied in a numerical model and compared under the working conditions of our experiment (Fig. 7).

Fig. 7 shows the effect of the interphase convection coefficient  $h_v$  on the temperature distribution of the solid  $T_s$  and fluid  $T_f$  phases along the axis of the receiver and the comparison between the experimental and numerical results. As can be observed, the correlations H4 apparently overestimate  $h_v$ , and the results of  $T_s$  are

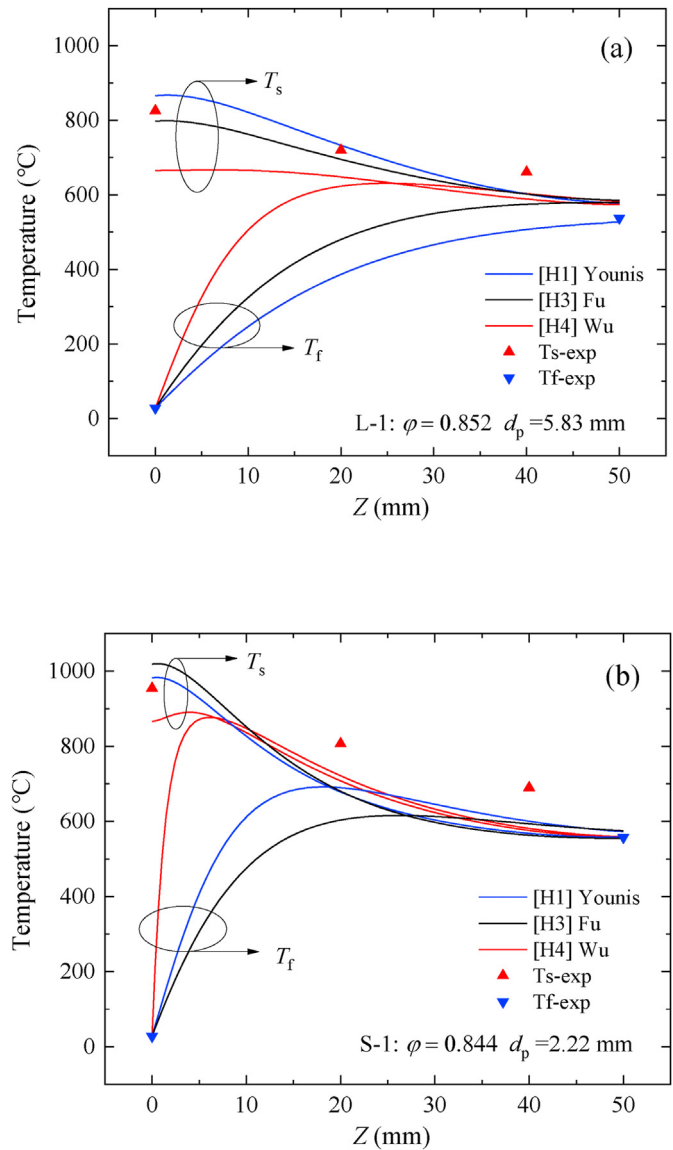


Fig. 7. Effect of the interphase convection coefficient  $h_v$  on the temperature distribution of the solid and fluid phases along the axis of the receiver, and comparison between the experimental and numerical results (a. L-1 receiver, b. S-1 receiver).

quite less than the experimental data. The correlations H3 apparently overestimate  $h_v$  in the case of L-1 (Fig. 7a), and results of  $T_f$  are slightly greater than the experimental data, but for the case of S-1 (Fig. 7b), it apparently underestimates  $h_v$ , and results of  $T_s$  are considerably greater than the experimental data at the inlet. Combining these two numerical cases, the results calculated by using the H1 correlations are in good agreement with the experimental data; therefore, H1 correlations are used in our numerical model.

### 5.3. Effect of air flow rate on fluid flow nonuniformity

Fig. 8 shows the spatial distribution of the fluid velocity ( $u_f$ ) of the absorber at different flow rates  $m_v$ . From the figure,  $u_f$  gradually increases along the flow direction, and the flow velocity gradient gradually decreases along the flow direction (the interval of velocity contours gradually increases along the flow direction) because the fluid is rapidly heated after entering the porous



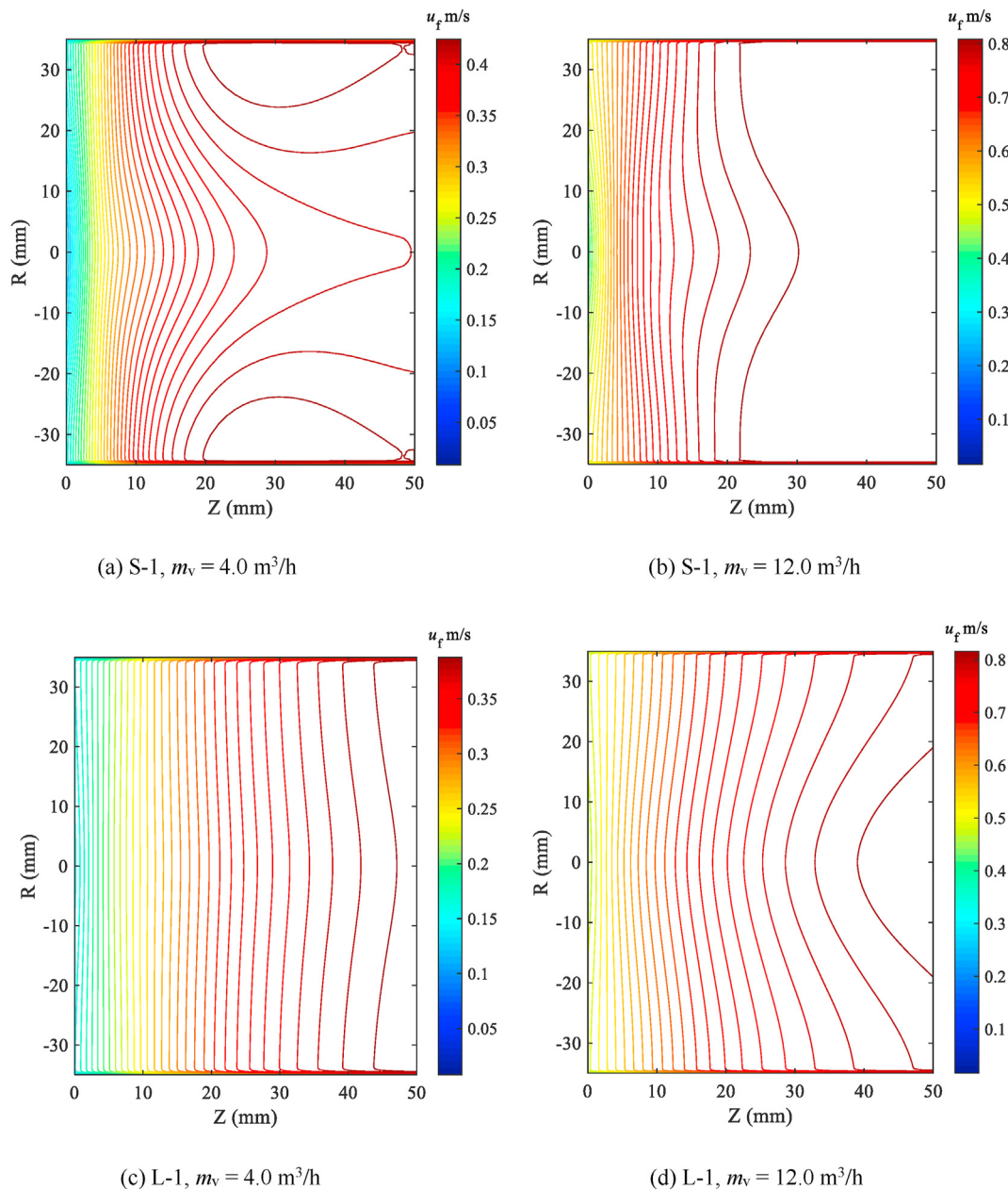


Fig. 8. Spatial distribution of the fluid velocity within the absorber at different flow rates ( $\phi \sim 0.85$ ).

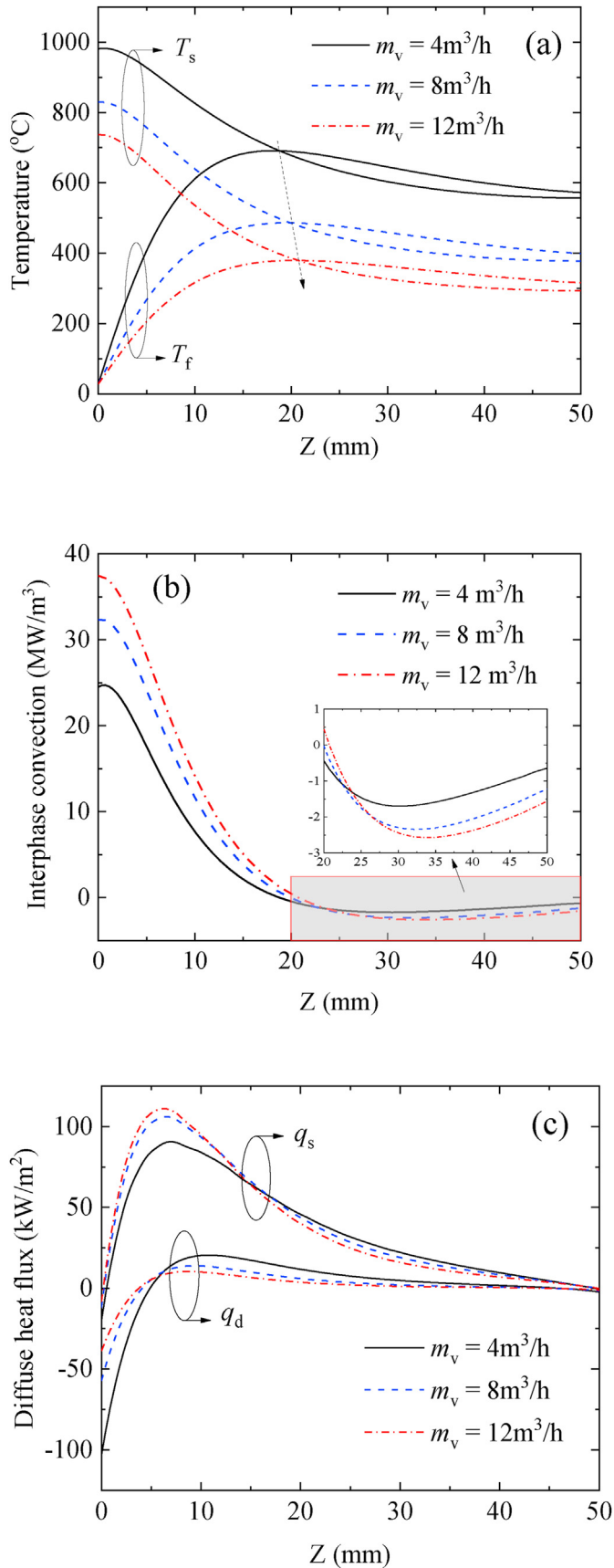
absorber, and the fluid density decreases with the increase in the temperature, leading to an increase in the velocity in the flow direction.

When  $Z > 10 \text{ mm}$ ,  $u_f$  exhibits a low distribution characteristic at the central region and a high distribution characteristic at the boundary wall (Fig. 8a and b). This result is attributed to the higher viscosity of the fluid at the central region with increase in the temperature, followed by the higher flow resistance in the high-temperature region. However, with the increase in the pore size  $d_p$ , the effect of viscosity changes due to the flow rate is reduced. At a small  $m_v$ ,  $u_f$  in the central region is only slightly less than  $u_f$  near the boundary wall (Fig. 8c and d); at a high  $m_v$  (Fig. 8d),  $u_f$  in the tail region is diametrically opposite because the higher the  $d_p$ , the larger the fluid channel in the porous medium, thereby weakening the effect of the fluid viscosity on the velocity. In this case, the

density change caused by temperature nonuniformity is the main reason that affects the velocity distribution.

Fig. 9 shows the influence of the flow rate  $m_v$  on the solid temperature  $T_s$ , the fluid temperature  $T_f$ , the interphase convective heat flux  $q_c$ , the conductive heat flux  $q_s$ , and diffuse radiant flux  $q_d$  on the central axis of the porous absorber.  $T_s$  continuously decreases in the flow direction, and  $T_f$  rapidly increases after entering the heat absorber, and it reaches a maximum when  $T_f$  is equal to  $T_s$  at an intersection point (Fig. 9a). With the increase in  $m_v$ , this intersection point moves to a low temperature and a deep position, as shown by the dotted arrow in the figure. This is an extremely interesting phenomenon, which is not interpreted in detail in the previous study, and it is closely related to the characteristics of convective heat transfer between phases.

Fig. 9 b shows the variation in the interphase convective heat



**Fig. 9.** Effect of the air flow rate  $m_v$  on the temperature, interphase convection, and conductive heat flux and diffuse radiant flux distributions along the axis of the absorber (a) and  $T_s$ ,  $T_f$ , (b)  $q_c$ , (c)  $q_s$ , and  $q_d$ . (case S-1:  $\phi = 0.844$ ,  $d_p = 2.22 \text{ mm}$ , POA = 1250 W).

flux  $q_c$  along the flow direction at different flow rates  $m_v$ . Along the flow direction,  $q_c$  drops to the lowest point and then increases. This result is consistent with the trend of the temperature difference between  $T_s$  and  $T_f$  (Fig. 9a): The higher the temperature difference, the higher is the  $q_c$ . With the increase in  $m_v$ , the peak value of  $q_c$  increases, indicating that the increase in the flow rate  $m_v$  leads to the considerable improvement in the convective heat-transfer capability.

Fig. 9 c shows the distribution of solid-phase conductive heat flux  $q_s$  and diffusion radiative flux  $q_d$  in the solid phase. Along the flow direction,  $q_s$  initially increases and then decreases to as low as zero. The essence of this variation is that the incident radiation is mainly absorbed in the vicinity of the inlet surface layer, leading to the maximum value for the solid temperature gradient. In addition, with the increase in  $m_v$ , the maximum value of  $q_s$  increases because the increase in  $m_v$  leads to the rapid decrease in the solid temperature, leading to a high-temperature gradient along the flow direction. Similarly,  $q_d$  initially increases from the minimum to the maximum and then slowly drops to zero along the flow direction. Notably,  $q_d$  is negative at the inlet, indicative of radiative loss from the irradiated surface. With the increase in  $m_v$ , this radiative loss decreases due to the decrease in the solid temperature.

#### 5.4. Effect of pore structure on thermal performance

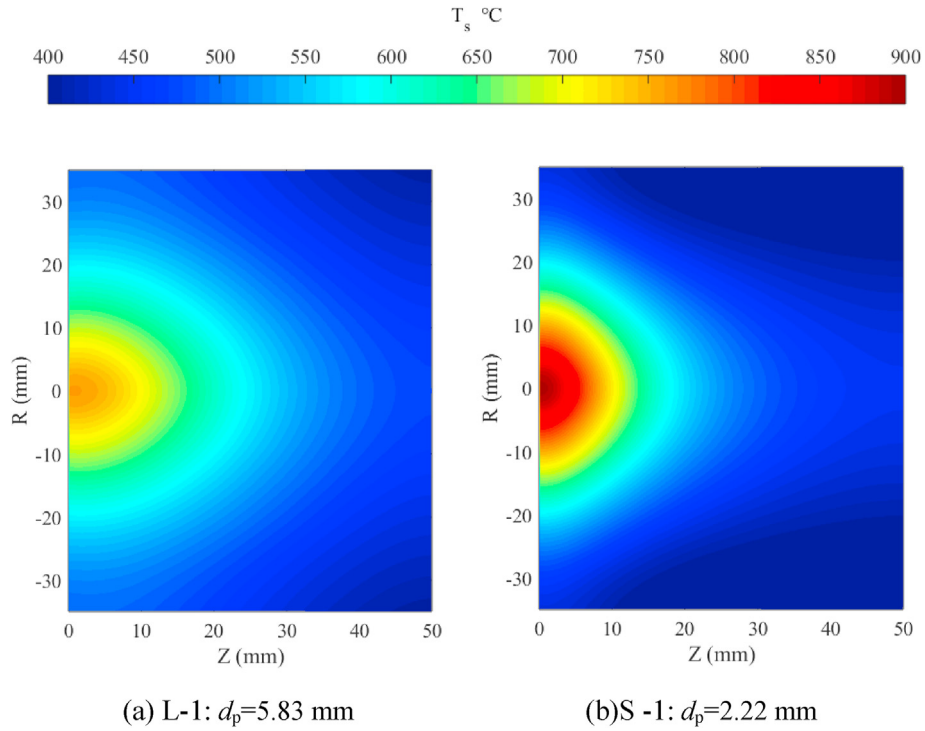
Porosity and pore size are important pore structure parameters of the porous medium, profoundly affecting the irradiation absorption, flow characteristics, and heat-transfer performance of the solar porous absorber. This chapter combines the experimental and simulation data of four porous absorbers to analyze the effect of porosity and pore size on the thermal and hydrodynamic behavior of solar porous absorbers.

Fig. 10 shows the effect of different pore sizes  $d_p$  on the solid temperature  $T_s$  distribution inside the absorber. In the case of  $d_p = 5.83 \text{ mm}$ , a temperature peak of  $748^{\circ}\text{C}$  occurs in the region near the inlet of the absorber, while in the case of  $d_p = 2.22 \text{ mm}$ , a considerably higher temperature peak of  $900^{\circ}\text{C}$  is observed because with the decrease in  $d_p$ , the peak value of solar radiative flux clearly increases (Fig. 3).

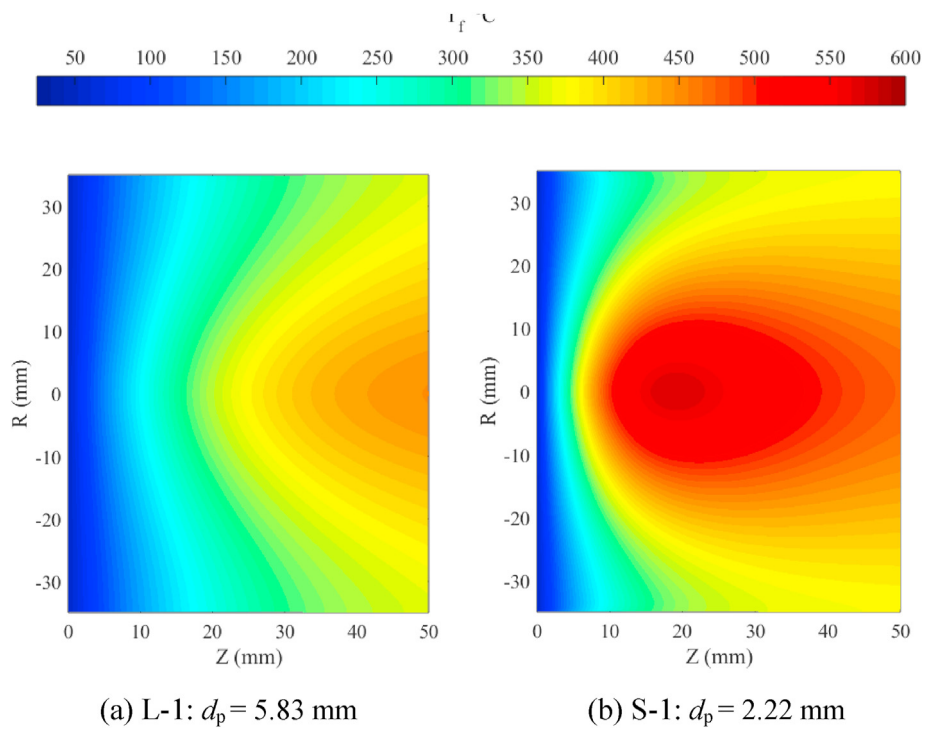
Fig. 11 shows the effect of the pore size  $d_p$  on the temperature distribution of the fluid temperature  $T_f$  inside the absorber. In the case of  $d_p = 5.83 \text{ mm}$ ,  $T_f$  gradually increases along the flow direction, the high-temperature area is mainly concentrated at the tail center, and the highest  $T_f$  is close to  $450^{\circ}\text{C}$ , while in the case of  $d_p = 2.22 \text{ mm}$ , a temperature peak of  $598^{\circ}\text{C}$  is observed in the central region of the absorber.

Fig. 12 shows the effect of the pore size  $d_p$  on the internal fluid velocity  $u_f$ . In the case of  $d_p = 5.83 \text{ mm}$ , the velocity gradually increases along the flow direction, and the velocity gradient is evenly distributed. In the case of  $d_p = 2.22 \text{ mm}$  (Fig. 12b), the flow velocity in the middle region is significantly less than that near the wall at the same  $Z$  position. This result is related to the fact that with the increase in  $T_f$ , the viscosity of the fluid increases, and the flow resistance in the region with high viscosity increases. In case of an absorber with a small  $d_p$  within narrower flow channels, the fluid flow is more significantly affected by the high nonuniformity of the temperature field.

Fig. 13 shows the effect of the pore size  $d_p$  on the solid temperature  $T_s$ , fluid temperature  $T_f$ , interphase convective heat flux  $q_c$ , conductive heat flux  $q_s$ , and diffuse radiation flux  $q_d$  on the central axis of the absorber. For the absorber with a small  $d_p$ ,  $T_f$  is slightly greater than  $T_s$  after increasing to the highest value (Fig. 13a). This inversion is related to the fact that with the decrease in  $d_p$ , the convective heat-transfer coefficient is considerably enhanced (Fig. 13b), and the solid region of the tail at a low temperature is



**Fig. 10.** Effect of the pore size  $d_p$  on the spatial distribution of the solid temperature ( $m_v = 6.0$  m<sup>3</sup>/h,  $\phi \sim 0.85$ ).

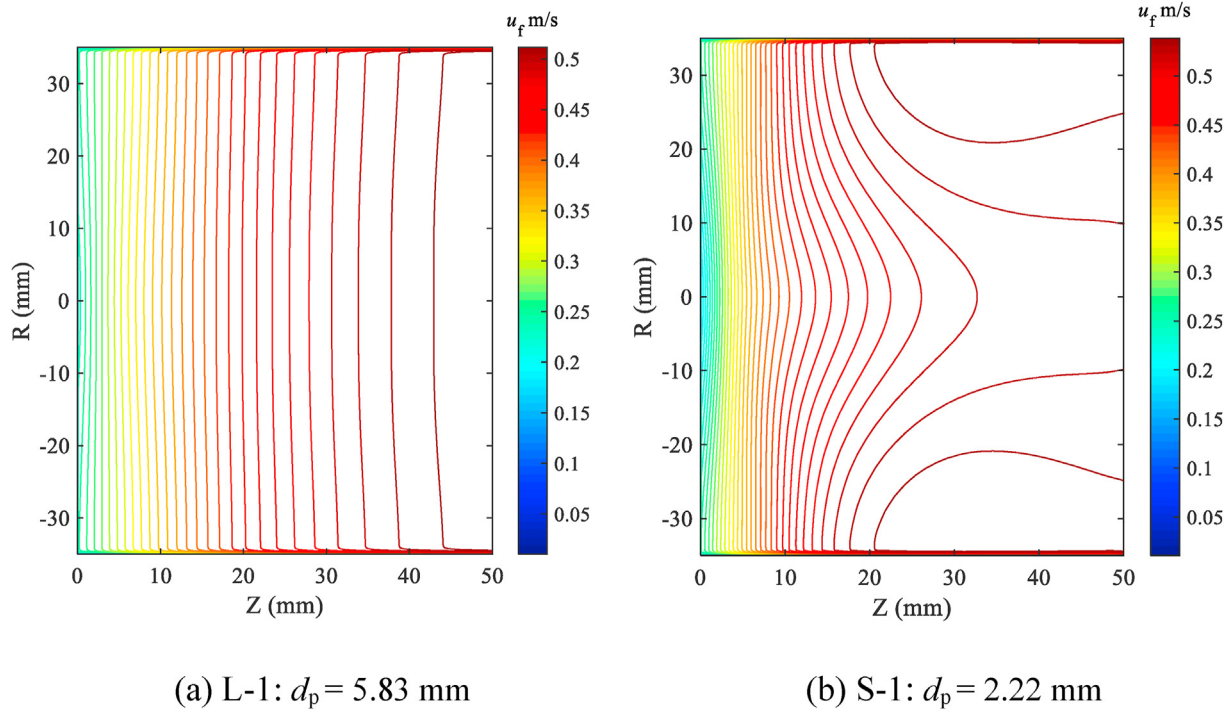


**Fig. 11.** Effect of the pore size on the spatial distribution of the fluid temperature ( $m_v = 6.0$  m<sup>3</sup>/h,  $\phi \sim 0.85$ ).

heated by the fluid that has attained an extremely high temperature in the high-temperature region. In addition, this phenomenon is observed in the simulation conducted by Barreto et al. [31] and Chen et al. [30]. However, for an absorber with low convection, this

inversion does not occur because the local thermal equilibrium is not attained. (Please check the case of  $d_p = 5.83$  mm in Fig. 13a).

Fig. 13 c shows the effect of the pore size  $d_p$  on the conductive heat flux  $q_s$  and diffuse radiation flux  $q_d$ . From the figure,  $q_s$  is



**Fig. 12.** Effect of the pore size on the spatial distribution of the fluid velocity ( $m_v = 6.0 \text{ m}^3/\text{h}$ ,  $\phi \sim 0.85$ ).

slightly less than zero at the initial position, indicative of the radiative heat loss at the inlet surface (the boundary condition of the solid-phase equation), while  $q_s$  is greater than zero, indicating that the direction of heat conduction is consistent with the direction of flow. In addition, with the decrease in  $d_p$ , the peak value of  $q_s$  increases because the gradient of  $T_s$  increases with the decrease in  $d_p$  (Fig. 13a). In addition, in the flow direction,  $q_d$  gradually increases from a negative value to a positive value, reaching the maximum value, and then decreasing to zero, and the absolute value of  $q_d$  increases with the decrease in  $d_p$  (Fig. 13c). These results indicated that a small pore structure is not conducive to radiation transmission.

Fig. 14 shows the effect of the porosity  $\phi$  on the solid temperature  $T_s$  distribution inside the porous absorber. In the case of  $\phi = 0.75$ , a temperature peak of  $900^\circ\text{C}$  occurs in the inlet region, while in the case of  $\phi = 0.95$ , a considerably lower peak  $T_s$  of  $705^\circ\text{C}$  is observed, indicating that the peak value of  $T_s$  decreases with the increase in  $\phi$  because with the increase in  $\phi$ , the extinction coefficient  $\beta$  decreases, resulting in the decreased interception of the incident rays.

Fig. 15 shows the effect of the porosity  $\phi$  on the temperature distribution of the fluid  $T_s$  inside the porous absorber. There is a peak value of  $T_f$  inside the porous absorber, and it increases with the increase in  $\phi$ . In the case of  $\phi = 0.75$ , the maximum  $T_s$  is observed in the central region at a depth  $Z$  of 20 mm, while in the case of  $\phi = 0.95$ , a deeper position of  $Z = 30 \text{ mm}$  is observed. This result is related to the higher porosity of the absorber, permitting the solar radiation to transfer deeper into its volume.

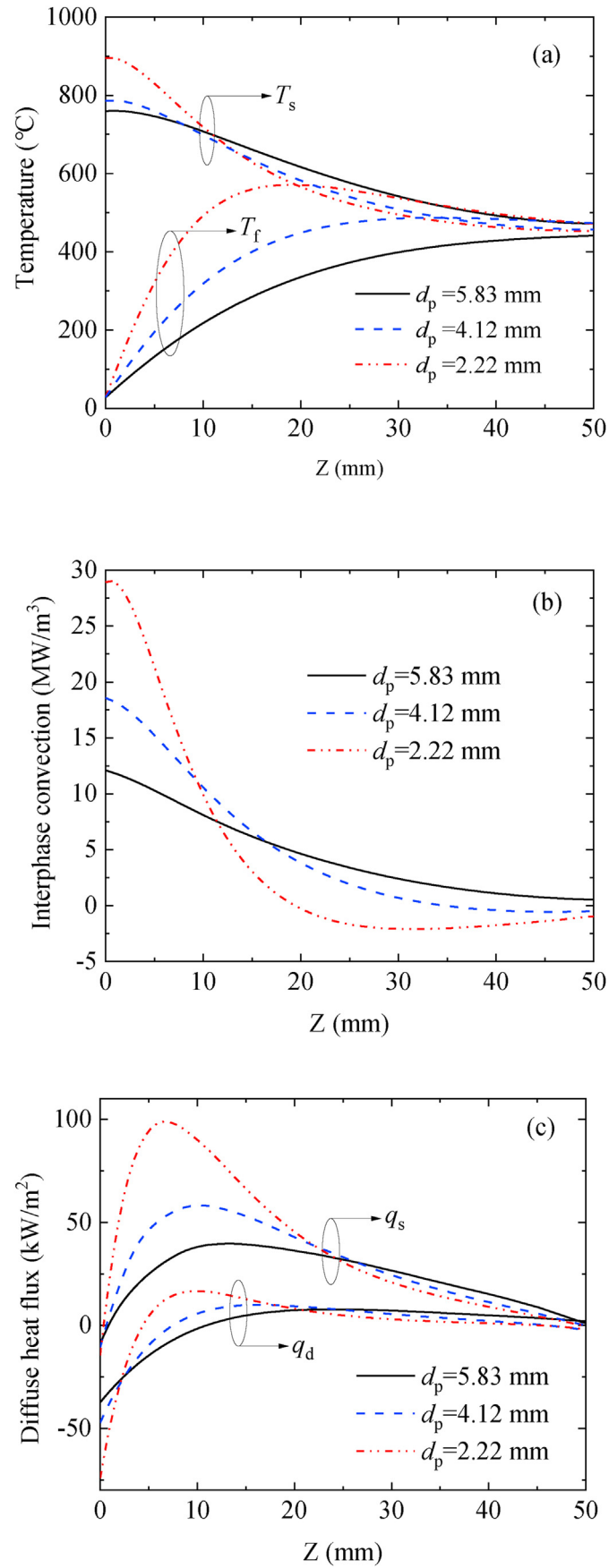
Fig. 16 shows the effect of the porosity  $\phi$  on the spatial distribution of the fluid velocity  $u_f$ . For both cases, the flow velocity in the middle region is less than that near the wall at the same  $Z$  position,

which is related to the fact that viscosity increases with the increase of the fluid temperature  $T_f$ , leading to the increase in the flow resistance.

With the increase in  $\phi$  from 0.75 to 0.95, the  $T_s$  at the inlet first increases and then decreases (Fig. 17a). This trend is mainly caused by two mechanistic actions due to changes in porosity. The lower the  $\phi$ , the higher the radiative heat flux on the irradiated surface, leading to high  $T_s$ ; however, the decrease in  $\phi$  leads to an increase in the pore velocity  $V_{\text{pore}}$  (at the same mass flow rate  $m_v$ ), enhancing the interphase convection  $h_v$  and eventually decreasing  $T_s$ . With the decrease in  $\phi$ , the convection heat flux at the inlet significantly increases (Fig. 17b). Fig. 17 c shows the effect of the porosity  $\phi$  on the conductive heat flux  $q_s$  and diffuse radiation flux  $q_d$  along the flow direction. With the increase in  $\phi$ , the peak value of  $q_s$  significantly decreases. Similarly, with the increase in  $\phi$ , the absolute value of  $q_d$  at the inlet decreases, indicating that the increase in the porosity  $\phi$  positively affects the reduction of radiation loss.

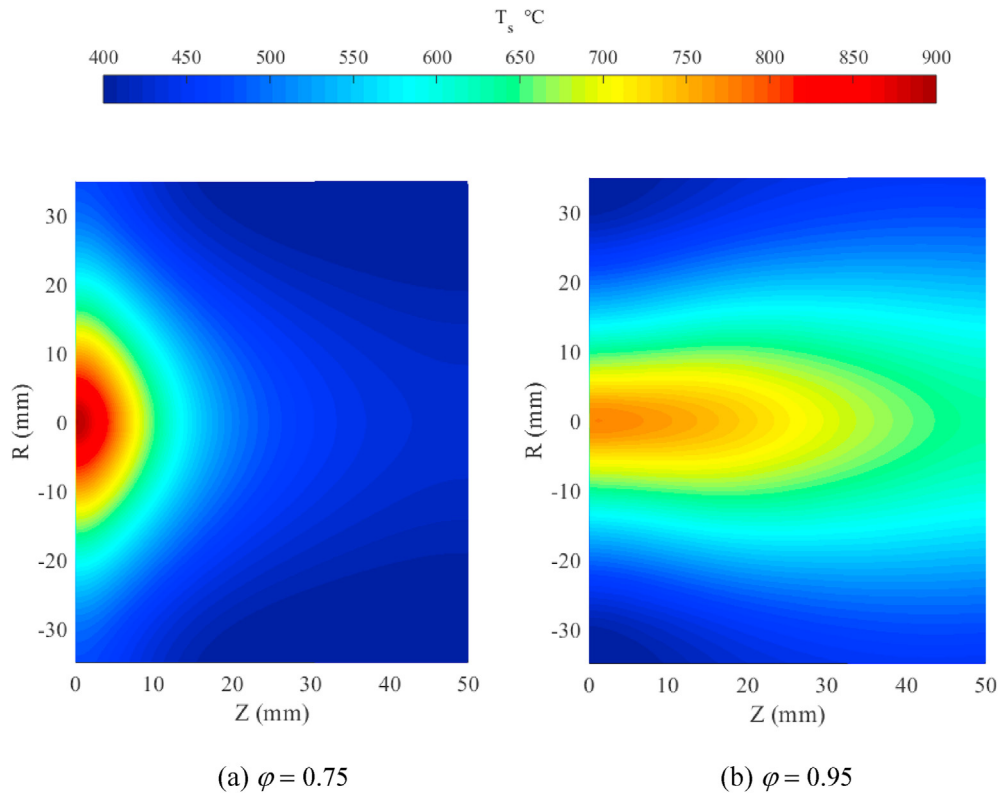
Table 4 concludes the experimental and the corresponding simulation data of different porous absorbers, including the average outlet temperature  $T_{f,o}$ , thermal efficiency  $\eta$ , and pressure drop  $\Delta p$ . The comparison of case L-1 and case S-1 (or L-2 and S-2) revealed that with the decrease in the pore size  $d_p$  at a constant porosity  $\phi$  leads to the increase in  $T_{f,o}$ , and hence higher  $\eta$  and higher  $\Delta p$ . With the decrease in  $d_p$  from 5.83 mm to 2.22 mm,  $T_{f,o}$  increases  $17.2^\circ\text{C}$ , leading to a nearly 3% increase in  $\eta$ . The decrease in  $d_p$  leads to the considerable improvement in the convective heat transfer, while for the small channel,  $\Delta p$  significant increases.

The comparison of S-0, S-1, and S-2 revealed that with the increase in  $\phi$  from 0.754 to 0.950,  $T_{f,o}$  and  $\eta$  first decrease and then increase, while  $\Delta p$  decreases. The above analysis revealed that the decrease in  $\phi$  would lead to an increase in the pore velocity  $V_{\text{pore}}$  (at

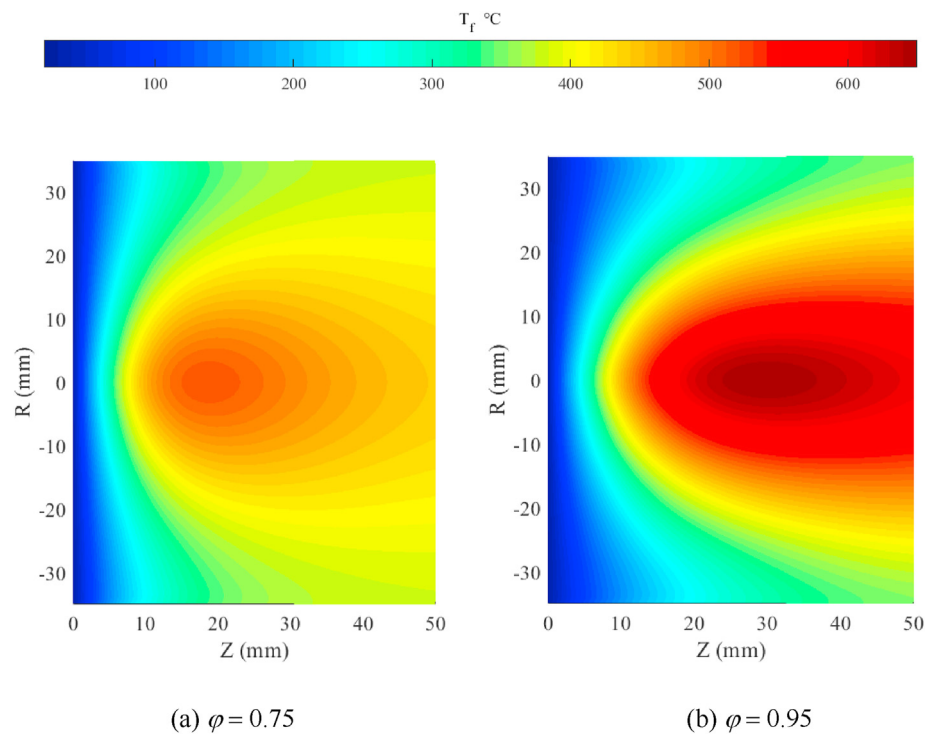


**Fig. 13.** Effect of the pore size  $d_p$  on the temperature, interphase convection, and conductive heat flux and diffuse radiation flux distributions along the axis of the absorber: (a)  $T_s$  and  $T_f$ , (b)  $q_c$ , (c)  $q_s$ , and  $q_d$ . (POA = 1250 W,  $m_v = 6.0 \text{ m}^3/\text{h}$ ,  $\phi \sim 0.85$ ).





**Fig. 14.** Effect of the porosity  $\phi$  on the spatial distribution of the solid temperature. ( $d_p = 2.22$  mm,  $m_v = 6.0$  m<sup>3</sup>/h).



**Fig. 15.** Effect of porosity on the spatial distribution of the fluid temperature ( $d_p = 2.22$  mm,  $m_v = 6.0$  m<sup>3</sup>/h).

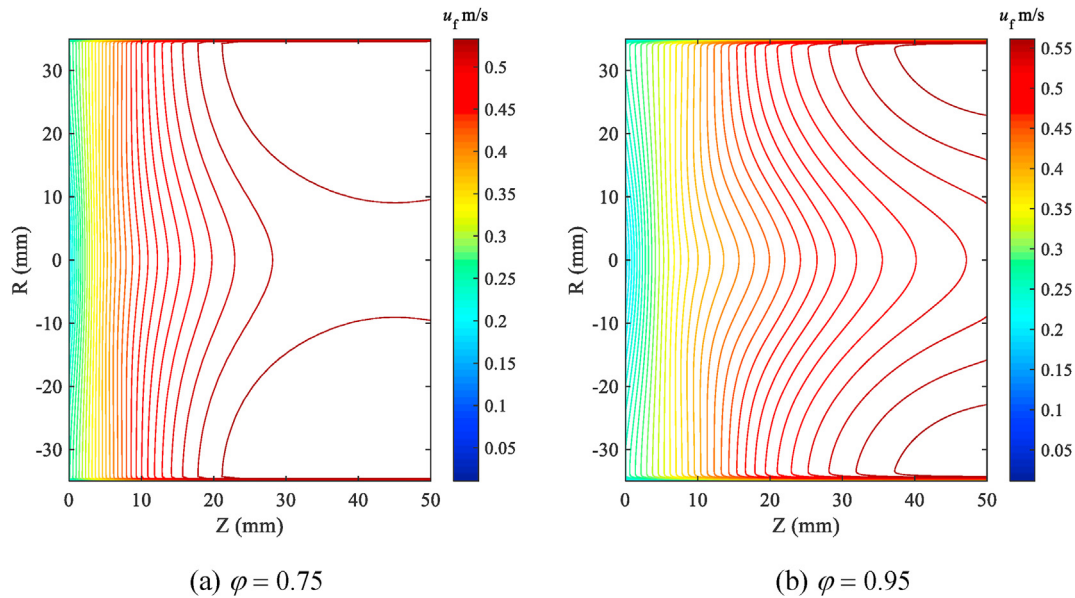


Fig. 16. Effect of the porosity on the spatial distribution of the fluid velocity. ( $d_p = 2.22$  mm,  $m_v = 6.0$  m<sup>3</sup>/h).

the same mass flow rate  $m_v$ ), thereby enhancing the interphase convection  $h_v$ . However, at a sufficient high  $\phi$ , another advantage becomes apparent: the higher the  $\phi$ , the deeper the solar rays penetrate the volume, thereby reducing the radiative heat flux on the irradiated surface.

The above results indicated that the porosity  $\phi$  and pore size  $d_p$  exert different effects on the thermal and hydrodynamic behavior of the porous absorber. Porosity  $\phi$  mainly affects the distribution of the radiative heat flux inside the absorber. The higher the  $\phi$ , the greater the probability of the light penetrating into the volume, and the lower the temperature of the irradiated surface, thereby reducing the radiation loss. The pore size  $d_p$  mainly affects the convective heat transfer between the solid and fluid phases. The smaller the  $d_p$ , the smaller the flow channel and the higher the tortuosity inside the porous medium, and the more violent the disturbance of the thermal boundary layer, thereby enhancing the heat exchange. Therefore, as a porous medium, an absorber with a high porosity and small pore size exhibits two major advantages: a more uniform radiative heat flux and a more effective thermal convective. These two advantages are essential for achieving a high thermal efficiency.

## 6. Conclusion

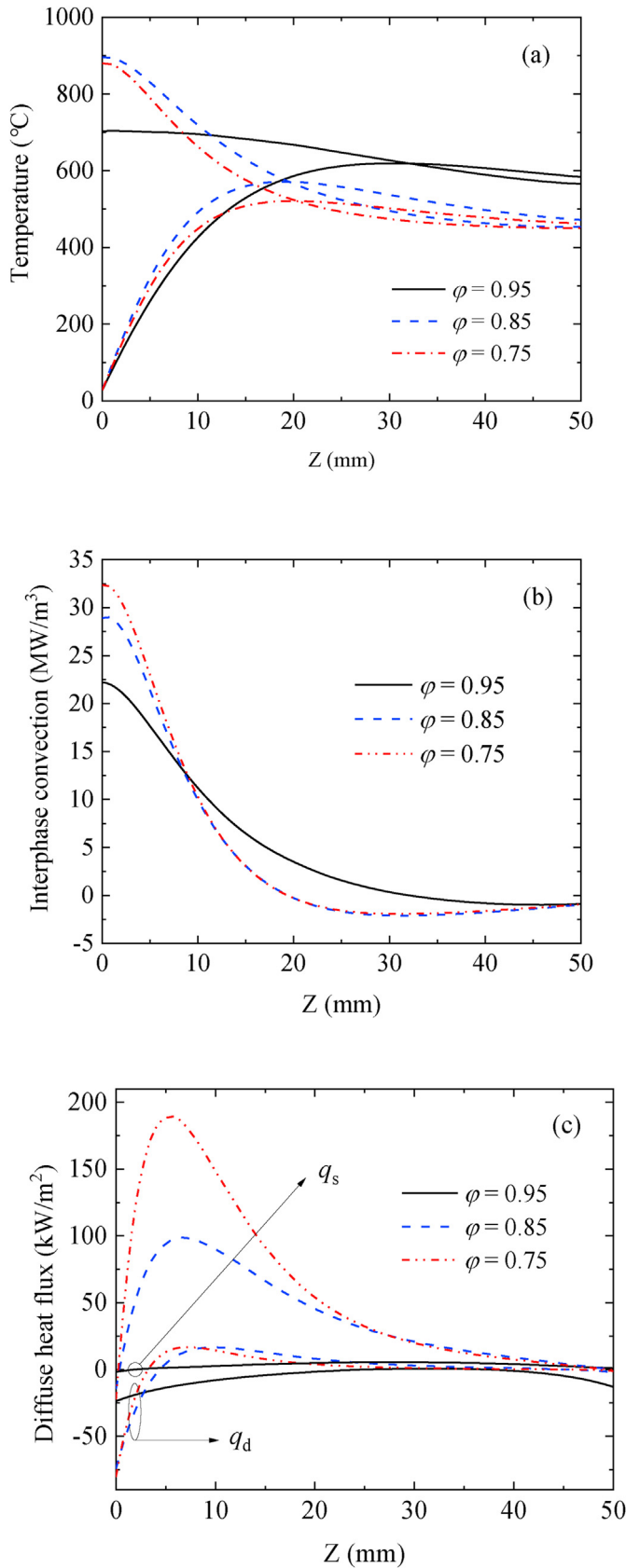
In this paper, experiments and numerical studies on the thermal and hydrodynamic behavior of solar porous volumetric receivers were conducted. Four SiC foams with various geometric parameters were investigated on a laboratory-scale test platform. Meanwhile, a three-dimensional theoretical model coupling the fluid flow and internal heat transfer was established. The solar ray transport from a xenon lamp to the volume of the absorber was solved by the Monte Carlo ray tracing (MCRT) method, and the obtained radiative heat flux is coupled with the local thermal non-equilibrium (LTNE) model to solve the internal heat transfer. Model validation was

achieved by comparing the experimental data and simulation results. The non-uniform and volumetric effect on the thermal and hydrodynamic characteristic in the absorber were presented. The key design parameters, such as porosity  $\phi$ , pore diameter  $d_p$ , volumetric heat-transfer coefficient  $h_v$ , and air flow rate  $m_v$ , were systematically analyzed. The following conclusions were obtained:

- (1) The velocity decrease caused by overheating in the center of the absorber was observed inside the S-1 absorber. Generally, the velocity of a fluid increased with the increase of temperature. This is caused by the decrease of the air density with the increase in temperature. However, at extremely high temperatures, fluid viscosity sharply increased, thereby increasing the flow resistance that forces the decrease in the flow rate.
- (2) The highest porosity  $\phi = 0.95$  case exhibited the highest thermal efficiency of 72.48%. The higher the  $\phi$ , the greater of the volumetric effect. On the other hand, the decrease in  $\phi$  would lead to an increase in the pore velocity  $V_{pore}$ , thereby enhancing the interphase convection  $h_v$ . This result explained the better performance for the  $\phi = 0.754$  case as compared with the  $\phi = 0.844$  case.
- (3) For absorbers with similar porosities ( $\phi \sim 0.85$ ),  $d_p$  decreased from 5.83 mm to 2.22 mm, the  $\eta$  increased by nearly 3%. For a small channel inside the porous absorber, the specific surface area for the contact between the cold fluid and the high-temperature solid phase increased, thereby enhancing the interphase convection.

## Credit author statement

P. Wang: Writing – original draft, Conceptualization, Methodology, Funding acquisition. J.B. Li: Software, Resources, Data



**Fig. 17.** Effect of the porosity  $\phi$  on the temperature, interphase convection, and diffuse heat flux distributions along the axis of the absorber: (a)  $T_s$  and  $T_r$ , (b)  $q_c$ , (c)  $q_s$ , and  $q_d$ . (POA = 1250 W,  $d_p$  = 2.22 mm,  $m_v$  = 6.0 m<sup>3</sup>/h).

**Table 4**

Comparison of the thermal and hydrodynamic performance between experimental data and numerical results.

| Case ID | $\phi$ | $d_p$ | $T_{f,o}$ exp.(°C) | $T_{f,o}$ sim.(°C) | $\eta$ sim.(%) | $\Delta p$ (Pa) |
|---------|--------|-------|--------------------|--------------------|----------------|-----------------|
| L-1     | 0.852  | 5.83  | 384.4              | 380.51             | 63.64          | 36.43           |
| L-2     | 0.770  | 5.67  | 398.6              | 384.14             | 64.43          | 49.82           |
| S-0     | 0.950  | 2.22  | —                  | 434.21             | 72.48          | 97.42           |
| S-1     | 0.844  | 2.22  | 401.6              | 407.95             | 68.65          | 109.31          |
| S-2     | 0.754  | 2.30  | 417.8              | 413.11             | 69.73          | 138.93          |

curation, Visualization. R.N. Xu: Investigation, Project administration. P.X. Jiang: Supervision, Writing- Reviewing and Editing, Funding acquisition.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Acknowledgment

The support of the National key research and development program (2018YFB1501003), Fundamental Research Fund for Central Universities (B200202172), and the “111” Project of Renewable Energy and Smart Grid (B14022) is acknowledged.

### Nomenclature

#### Abbreviations and subscripts

|       |                                                      |
|-------|------------------------------------------------------|
| $C_p$ | Specific heat of fluid at constant pressure [J/kg/K] |
| $d_p$ | Pore diameter [m]                                    |
| $L$   | Length [m]                                           |
| $N$   | Number                                               |
| $P$   | Power [W]                                            |
| POA   | Power over the receiver aperture [kw]                |
| $Q$   | Heat source term                                     |
| Re    | Reynolds number                                      |
| SiC   | Silicon Carbide                                      |
| $T$   | Temperature [°C, K]                                  |
| $V$   | Volume [m <sup>3</sup> ]                             |

#### Greek symbols

|           |                                                 |
|-----------|-------------------------------------------------|
| $\beta$   | Extinction coefficient                          |
| $\chi$    | temperature unevenness                          |
| $D$       | Diameter                                        |
| $F$       | Fow resistance                                  |
| $h$       | Heat transfer coefficient [W/m <sup>2</sup> /K] |
| $\phi$    | Porosity                                        |
| $K$       | Permeability tensor [m <sup>2</sup> ]           |
| $\lambda$ | Thermal conductivity [W/m/K]                    |
| $M$       | Mass flow rate [kg/s]                           |
| $\mu$     | Viscosity [Pa/s]                                |
| $\theta$  | Polar angles                                    |
| $\rho$    | Density [kg/m <sup>3</sup> ]                    |
| $u$       | Velocity [m <sup>2</sup> /s]                    |
| $\xi$     | Random number                                   |
| $\psi$    | Azimuth angles                                  |
| $\eta$    | Efficiency [%]                                  |
| $\kappa$  | Absorption coefficient                          |
| $\sigma$  | Scattering coefficient                          |

## Subscripts

|     |             |
|-----|-------------|
| abs | Absorbed    |
| e   | Environment |
| f   | Fluid       |
| inc | Incident    |
| p   | Porous      |
| r   | Receiver    |
| s   | Solid       |
| sca | Scattering  |

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